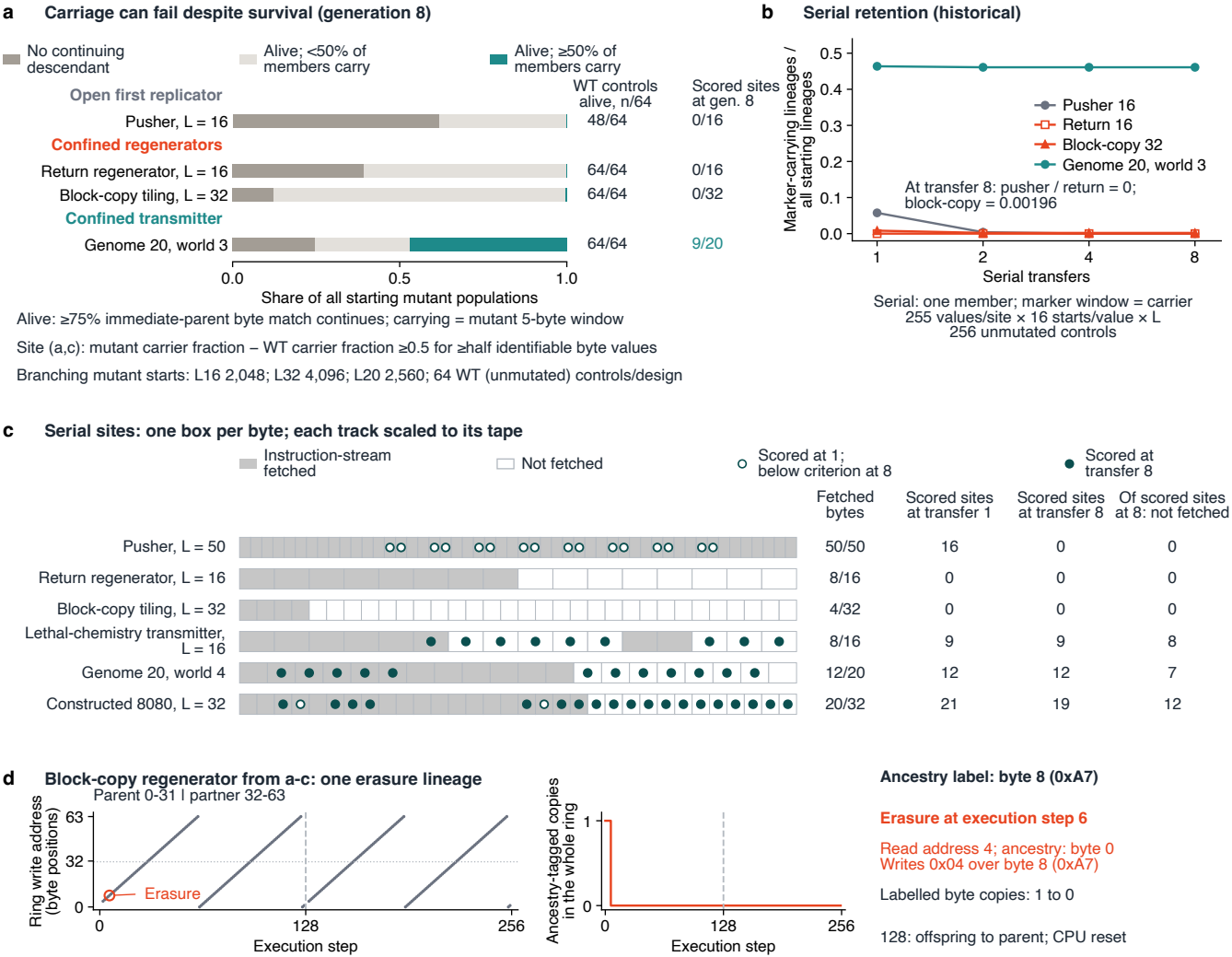
a
Carriage can fail despite survival (generation 8)
No continuing descendant
Alive; <50% of members carry
Alive; ≥50% of members carry
WT controls alive, n/64
Scored sites at gen. 8
Open first replicator
Pusher, L = 16
48/64
0/16
Confined regenerators
Return regenerator, L = 16
64/64
0/16
Block-copy tiling, L = 32
64/64
0/32
Confined transmitter
Genome 20, world 3
64/64
9/20
0.0
0.5
1.0
Share of all starting mutant populations
Alive: ≥75% immediate-parent byte match continues; carrying = mutant 5-byte window
Site (a,c): mutant carrier fraction – WT carrier fraction ≥0.5 for ≥half identifiable byte values
Branching mutant starts: L16 2,048; L32 4,096; L20 2,560; 64 WT (unmutated) controls/design
b
Serial retention (historical)
Marker-carrying lineages / all starting lineages
0.5
0.4
0.3
0.2
0.1
0.0
Pusher 16
Return 16
Block-copy 32
Genome 20, world 3
At transfer 8: pusher / return = 0; block-copy = 0.00196
1
2
4
8
Serial transfers
Serial: one member; marker window = carrier
255 values/site × 16 starts/value × L
256 unmutated controls
c
Serial sites: one box per byte; each track scaled to its tape
Instruction-stream fetched
Not fetched
Scored at 1; below criterion at 8
Scored at transfer 8
Fetched bytes
Scored sites at transfer 1
Scored sites at transfer 8
Of scored sites at 8: not fetched
Pusher, L = 50
50/50
16
0
0
Return regenerator, L = 16
8/16
0
0
0
Block-copy tiling, L = 32
4/32
0
0
0
Lethal-chemistry transmitter, L = 16
8/16
9
9
8
Genome 20, world 4
12/20
12
12
7
Constructed 8080, L = 32
20/32
21
19
12
d
Block-copy regenerator from a-c: one erasure lineage
Parent 0-31 | partner 32-63
Ring write address (byte positions)
63
32
0
Erasure
0
128
256
Execution step
Ancestry-tagged copies in the whole ring
1
0
0
128
256
Execution step
Ancestry label: byte 8 (0xA7)
Erasure at execution step 6
Read address 4; ancestry: byte 0
Writes 0x04 over byte 8 (0xA7)
Labelled byte copies: 1 to 0
128: offspring to parent; CPU reset